

Specifications

In this exercise, you will have to write a small tool to manage a list of reports. The application will allow the user to:

- Save report files and associate it with the name of the author;
- Fetch the list of authors and list of reports for each author;
- Save a copy of a report on the disk.

The tool will use a command-line interface that should be close to the following example:

```
$ run-report-management
Welcome in report management system!
Loading database... done.
Please enter a command:
> add, reports/report-1.md, Elena Voss
> add, reports/report-2.md, Elena Voss
> add, reports/report-3.md, Marcus Hale
> add, reports/report-4.md, Elena Voss
> add, reports/report-5.md, Marcus Hale
> add, reports/report-6.md, James Reynolds
> authors
Elena Voss
James Reynolds
Marcus Hale
> reports, Marcus Hale
report-3.md
report-5.md
> save, James Reynolds, report-6.md, temp/report-copy.md
Report successfully copied to temp/report-copy.md
> exit
```

The program that you will produce to create this application needs to fulfil the following constraints:

- The programming language used will be one of the following: C++, Python, Javascript, or Dart.
- The program is intended for offline use on the client computer. No internet access should be required to use it, and no big dependency should be included (no MySQL or PostgreSQL for example).
- The command execution should be reasonably fast once the program has been opened, even for large sets of reports.

For the sake of this exercise, it can be assumed that the user input will always be correct (no typo, no wrong commands). There will be then no need to sanitize the user input. The program

is not supposed to read the report files, only to save them.

Deliverables

To complete this assignment, you are expected to provide the following:

1. The source code of your program, with the associated build files when applicable.
2. A diagram or short documentation explaining the architecture of the program.